system disk or the ones currently in use in memory.

When the settings appear on the screen, select the programmable function keys by typing E. Then, for each function key number from zero to nine, type the number and then the definition shown below:

```
0:^OR52↵  (Set the right margin at 52 columns.)
  ^OH     (Turn off the hyphen help.)
  ^JHO    (Turn the Help level to off.)
  ^OJ     (Turn off the right margin justification.)
1:^KB     (Start a block of text to be moved.)
2:^KK     (End a block of text to be moved.)
3:^KV     (Move the block of text.)
4:^B      (Reform a paragraph.)
5:^KD     (Exit and save text.)
6:^Z      (Read ahead.)
7:^-      (Delete to left of cursor.)
8:^G      (Delete at cursor.)
9:^W      (Read back.)
```

When you are finished defining these function keys, press Return to go back to the settings menu, then Return again to go to the save menu. Insert the WordStar disk in the B: drive, and type B to save the modified SETUP configuration to the B: drive.

That's it! Now your WordStar disk can be used by novices following the instructions presented above. I hope this article helps you or your trainees become more productive with WordStar.

Diane Paige Thompson, Ph.D., is a professor at the Northern Virginia Community College.

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Metal Messages From The Oracle

We appreciate the favorable and generally accurate coverage of our bulletin board system in the June/July 1984 issue of *The Portable Companion*. I would, however, like to make a few corrections.

First, Tim Gary is the sole author of the bulletin board system, although he and I are joint authors of the accompanying documentation.

Secondly, the name of the product is now Metal Message System, and that name is a trademark of our company, Delphi Data Systems.

Thank you again for the review and for your fine publication.

Byron McKay
Mountain View, CA

Reverse Video & Block Cursor Fix

Many thanks for bringing *The Portable Companion* back to life and providing us with excellent information. My son, James H. Lovell, and I were delighted with Thom Hogan's [article about having a] block cursor, but a bit disappointed to find it only functioned on 52 columns (June/July 1984 issue, p. 24).

I noticed letters from John A. Hansen and Jerry Boyce in the November issue, which I just received, indicating interest in getting help on the subject.

Never fear. James H. Lovell to the rescue!

He has found a "fix" so that the block cursor operates with Screen-Pac 80 and it can be carried out without removing the piggy-back board containing the Screen-Pac 80 electronics.

NOTE! Be careful of static. The HC type chip is a high speed CMOS.

1. Following Thom Hogan's excellent drawings in the July 1984 issue of *The Portable Companion*, pages 25 and 26, note the following changes:

2. The 74LS86 chip, is piggy-backed to chip U2 on the upgrade board. U2 is a 74HCOO chip and is the second chip from the left rear corner of the upgrade board. From the 74LS86 chip, cut off pins 1, 2, 3, 4, 5, 6, 11, 12, and 13. NOTE: THESE PINS ARE NOT THE SAME ONES LISTED IN THOM HOGAN'S ARTICLE!

3. Carefully cut pin 8 of U2 (74HCOO) at the board and on the chip.

4. Place chip 74LS86 on top of U2 (74HCOO) so that pin 7 of the piggy-back contacts pin 7 of U2. Solder these two pins.

5. Connect pin 8 of the piggy-back 74LS86 to the board where pin 8 of U2 was cut. You may be able to urge pin 8 into the board by applying heat (solder iron) to the board where pin 8 of U2 was connected.

6. Solder pins 9, 10 and 14 of 74LS86 to pins 9, 10 and 14 of U2 (74HCOO). In summary, pins 7, 9, 10 and 14 of 74LS86 are soldered to pins 7, 9, 10 and 14 of 74HCOO and pin 8 of 74LS86 replaces pin 8 of 74HCOO.

7. Remove pin 1 of U23, 74HC74, from the board either by softening the solder or cutting it at the board. U23 is the fourth chip from the right end of the board on the front edge and is immediately to the right of the cable header for the flat cable originating on the main logic board. Lift pin 1 away from the board. Solder to it a short length of wire-wrap wire, or other suitably insulated wire from which the insulations on the ends have been stripped. Carry the free end of this wire over the top of U23 and solder it to pin 14 of U23.

8. That's it. After assembling the computer, you should have a block cursor in reverse video that functions with the Screen-Pac 80 upgrade.

James S. Lovell
Charleston, WV

Drive C: Review

The review of the Drive C: unit in the April issue by Don Krantz is out to lunch.

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